

Detection of Black Hole Attacks in Mobile Ad Hoc Networks Using Optimization–Based Routing Algorithms

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Abstract:

Mobile Ad hoc NETWORKS (MANETs) are a new type of wireless communications that are operating in a highly dynamic and unpredictable environment. However, security is a major issue in routing, because of moving nodes, mainly because it suffers from a black hole attack. This research focuses on the detection of Black Hole attacks using advanced optimization algorithms, namely Grey Wolf Optimization (GWO), Ant Colony Optimization (ACO), Genetic Algorithm (GA), and Particle Swarm Optimization (PSO). This survey aims to enhance the performance of MANETs by effectively identifying and mitigating the impact of malicious nodes. In the suggested approach, the GWO algorithm optimizes the routing paths, improving throughput and packet delivery ratio while reducing end-to-end delay. The ACO utilized a structured mechanism for routing decisions, utilizing pheromone updates and heuristic values to enhance the efficiency of Black Hole attack detection. The GA framework integrates Crossover and Mutation operators, offering a more structured generation of nodes and improving adaptability to dynamic network conditions. Additionally, the PSO algorithm introduces a multidimensional approach, incorporating diversity optimization and energy-aware routing for Black Hole detection. Hence, the suggested technique enhances the packet transmission in MANET and secures the routing layer effectively.

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